

23

The project manager

Learning outcomes

When you have finished reading this chapter, you will be able to:

- Describe the role, satisfactions and challenges of the project manager's role
- Identify areas of your work that give you an opportunity to develop
- Summarize the British Computer Society's code of conduct
- Apply appropriate ethical standards to your work.

23.1 Introduction

You should by now have a clear idea of the job of the project manager, and you may be asking yourself, 'What sort of person makes a good project manager?' This chapter considers just that question. We have looked at it in a number of ways and set out some ideas for you to consider. We have included:

- One company's vision of the role of the project manager.
- An outside perspective showing what the IS project manager can learn from management in other fields.
- A developmental approach used in one company we have studied.
- How psychometrics can help in the selection of the right person for the job and with the development of the right person once they are in post.
- One company's research into how psychometrics helped.

Taken together, these approaches to the topic should give you a clear outline of what it is like to be a project manager in an IS environment. And it should also help you to clarify your own idea about your development as a project manager.

23.2 The vision

One major international systems company created a vision statement that, for them, summed up what they wanted their project managers to commit to.

They accepted that not all of it will be true for all of their project managers all of the time, but they have defined the feeling of the role in a way that everyone who starts managing projects for them knows what is expected of them in terms of attitude and final goal. As you read it, ask yourself if you would want to be seen like this by your manager, your peers, your customers and your team. If the answer is yes, then maybe it could be a vision for you as well as for others.

As a project manager, you are determined to succeed and to bring your project to a successful conclusion – on time, within budget and to the customer's satisfaction. This isn't easy but it is rewarding.

It's not easy. You have to meet the client's requirements and bring together equipment, software and people to deliver a solution to a problem that hasn't been solved before. You know that equipment will be late, that software will give you problems and that you won't always get the people you need when you need them.

You accept this. It's no use crying over spilt milk, you have to get on with the job: replan, reorganise, reschedule, reallocate and remotivate. You said that you didn't want a boring job and that you wanted a challenge, an opportunity to shape events, the chance to build your reputation and your company's – well, you got it. Overcoming problems and winning through to the end is part of what turns you on. It feels good to manage a complex issue through to a successful conclusion.

But you don't have to do this on your own. You are the leader, but it's a team task and a team success. Team spirit, helping people grow and harnessing their skills and enthusiasm is important to you. You have a personal commitment to their development. You manage people well as individuals and as a team. They respond to you and trust your judgement. You enable them to perform.

Managing projects is what you do best. People know you and you are valued. Each project is different, each a new challenge. You welcome this challenge: it gives you new opportunities to learn. Your contribution is consistently above average, beyond the norm – even in a goal-orientated, customer service industry like ours. You wouldn't have it any other way.

For us, this sums up the essence of being a project manager. Whilst all of the obstacles are perhaps not present in every project, the spirit of this vision statement is uplifting and motivational.

23.3 An outside perspective

Many project managers feel that their role is very different from that of the traditional line manager and thus that they can learn very little from old-style management textbooks. Let us examine this belief. If we define a project as something with a clear start and finish, as something that brings about change and that has a clearly stated objective, then humanity has been managing projects since the dawn of time. Hunting a woolly mammoth and raising a crop are both projects that conform to this description. Given that, are there many differences between the relatively new discipline of project management and that of older, traditional management? Traditional management priorities often centre on control: controlling the team, the task and the individual.

Classic texts tell us that the traditional manager has the functions of control, coordination, communication and the setting of performance standards. Are these so different from what today's project manager does?

There are some major differences. For example, there are differences in timing:

- Line management is continuous whereas a project is finite.
- Line managers expect to stay in post for a considerable length of time whereas project managers expect to manage a project for a finite length of time and then move on to a new and different challenge.
- Line managers have the opportunity to improve the performance of their operations gradually and incrementally. Project managers, on the other hand, usually only have one chance to get things right – or wrong – in a project before they have to move on to something else.

As well as this, the role of the IS project manager is to create a product – usually a new IS system incorporating hardware and software. The primary tool in this task is the brainpower of the project team. In this the project manager is different from traditional managers whose primary task is often to keep the status quo operational, whether this is to keep the assembly line moving or the office department functioning at full efficiency. In essence, traditional management is often seen as being repetitive whereas project management is mostly non-repetitive.

If project managers want to make best use of the skills and abilities of their highly able workforce, they must be willing to give them information so that they can develop a rational understanding of the tasks assigned to them. A skilled software engineer will function much better if he or she knows not just what is required but why it is required. In this way, the software engineer is not only doing the required job but is also quality checking the analysis and design work that was done previously. Again, this is different from traditional management.

Whereas traditional management often looks in great detail at how the workforce should be supervised – with close supervision often being recommended – many, if not all, professional engineers will look for an absence of detailed supervision, freedom from administrative routine where feasible and a working environment which sets them free to perform at their best. Naturally, in such an environment, the project manager has to trust the team to do a good job. The degree of control and supervision given so that they can do this good job will be different from many earlier management disciplines.

This shows that there is greater possibility for failure in project management than in traditional management. In traditional management, a manager will establish a closed loop in which staff performance is noted and measured, compared with performance standards and then any necessary corrective action is taken.

Particularly during the initial stages of an IS project, the analysis and design phases, the project manager is often flying blind. The degree of risk and uncertainty will vary from project to project, but all projects will contain some risk, and many are highly risky. Nothing exactly the same as this piece of work has

been attempted before, so the project manager often faces real difficulty – in spite of all the tools and techniques – in predicting the outcomes of the work being done, and has the additional problem of knowing that any mistakes will have profound and perhaps costly consequences later in the project.

If there are differences between traditional management and IS project management, are there similarities between general project management, which has been a recognised discipline for over fifty years, and IS project management? Consider the following extract from an article published in the *Harvard Business Review* in 1960. It is by Paul Gaddis and is a classic early discussion of the emerging role of project management in hi-tech industries:

Generally speaking, the project manager's business is to create a product – a piece of advanced-technology hardware. The primary tool available to him is the brainpower of men who are professional specialists in diverse fields. He uses this tool in all the phases of the creation of his product, from concept through the initial test operation and manufacturing stages.

The article then goes on to address many of the issues current today: what does the project manager do, what kind of person is he, what training is required? So, in over forty years of examination we are still puzzling over the same issues. Like leadership, perhaps project management is not the simplest job in the world. As an illustration of how the world keeps on reinventing the same old wheel, we came across a definition of project management and a list of the skills of a project manager produced quite recently by a well-respected international body.

A project was defined as 'a set of interrelated activities, with an agreed start and finish time, that is undertaken by an organization in order to meet defined objectives within the constraints of time, cost and resources at specified quality requirements', and a project manager as 'the individual to whom authority, responsibility and accountability have been assigned for the overall management of resources – including the technical, time and cost aspects – of a project, and the motivation of all of those involved'.

When looking at the project manager in this context, those skills that will make a good project manager together with the attributes and skills which the project manager should develop to ensure the effectiveness of the team and the individuals in that team were described. There is much here that reflects the work of Gaddis all those years ago, much that reflects the engineering disciplines and the contractual nature of the project manager's role, and much that is currently occupying present-day thinking about the development of project managers. These skills are:

- *Leadership.* Project managers should be able to stimulate action, progress and change.
- *Technological understanding.* Project managers need to have an accurate perception of the technical requirements of the project so that business needs are addressed and satisfied.
- *Evaluation and decision making.* Project managers should have the ability to evaluate alternatives and to make informed decisions.

- *People management.* Project managers should be able to motivate and enthuse their teams and have a constant personal drive towards achieving the project's goals.
- *Systems design and maintenance.* Project managers should be able to demonstrate their individual competence and have a complete working knowledge of the internal administration of their project.
- *Planning and control.* Project managers should be constantly monitoring progress against the plan and taking any necessary corrective action using modern planning and monitoring methods.
- *Financial awareness.* Project managers should be proficient in risk management and have a broad financial knowledge.
- *Procurement.* Project managers should understand the basics of procurement and be able to develop the procurement strategy for their project.
- *Communication.* Project managers should be able to express themselves clearly and unambiguously in speaking and writing and be able to do this in a wide range of situations and with a wide range of people.
- *Negotiating.* Project managers should be skilful in managing their clients and should be able to plan a negotiation strategy and then carry it out.
- *Contractual skills.* Project managers must be able to understand the contract that defines their project and should be able to manage subcontractors to ensure that the contractual terms are met.
- *Legal awareness.* The project manager should have an awareness of any legal issues that could affect the project.

23.4 A developmental approach

It is clear that in any project team the project manager is the key figure. However, as we have seen, although there is no such thing as a standard project in the IT industry, there are certain skills and qualities of project managers which can contribute materially to the success of any project with which they are associated. These skills and qualities are increasingly called competences, but they are often little more than the list above: groups of abilities and talents which shape the way the project manager does the job and so have an impact on the whole project. We can get a better grasp of these skills, abilities or competences if we break them down into four key areas. The skills and qualities that project managers use to manage:

- Themselves, such as time management.
- Their one-to-one interactions, such as negotiation skills.
- Their one-to-many interactions, such as with their team.
- The task, such as monitoring.

Let us look at them one by one, starting with self-management. Anyone setting out to manage the complexity of a project must be able to manage their own self. Many of the key qualities and skills for self-management are identified in Figure 23.1.

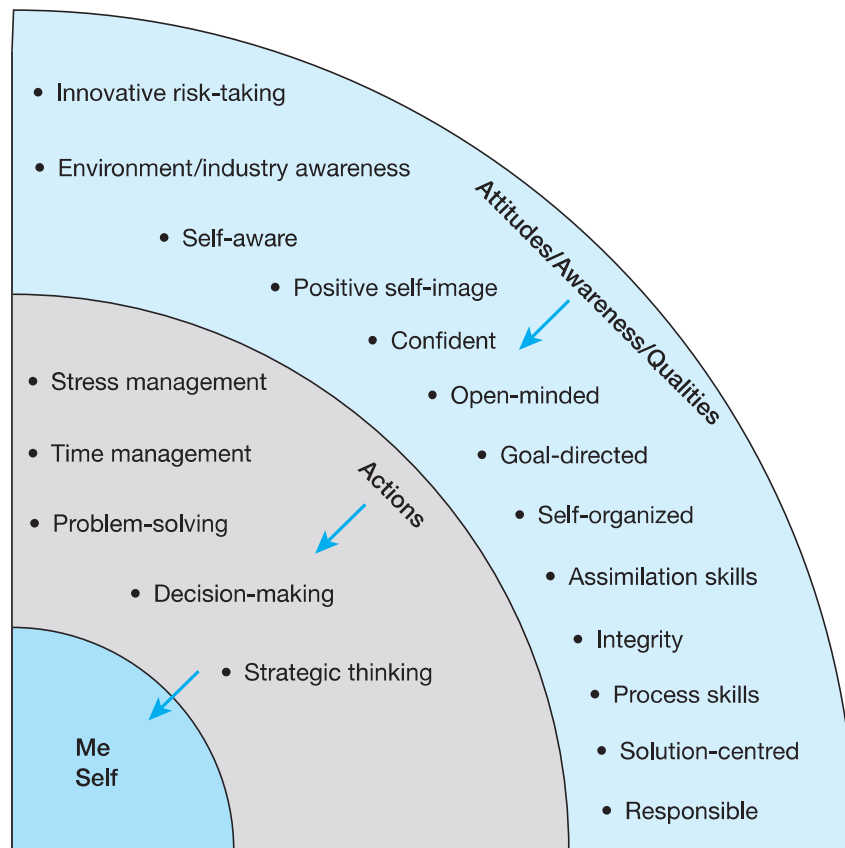


Figure 23.1 Self-management

In addition, it is expected that a project manager provides the strength of character and the vision to keep the project on course. Along with this would go the ability to be seen as a respected and commanding figure within the team, the company and the customer's organization. This will not come out of thin air, but it is often something that can be fostered in new project managers by their seniors. Without role models, it is unlikely that any aspiring project manager will ever learn what leadership looks like in their organization. The project manager will set the tone for all of the interactions on the project. In every interaction, those who are involved learn the way the project manager does things, and this then becomes the project way of doing things.

Dealing with people on a one-to-one basis is a major part of the project manager's job. Sometimes these interactions will be with customers, sometimes with subcontractors, sometimes with project members; sometimes they will be with more senior staff, sometimes with junior staff. Regardless of whom they interact with, project managers must be seen to be professional and competent by all who watch them. Many of the key qualities and skills are shown in Figure 23.2.

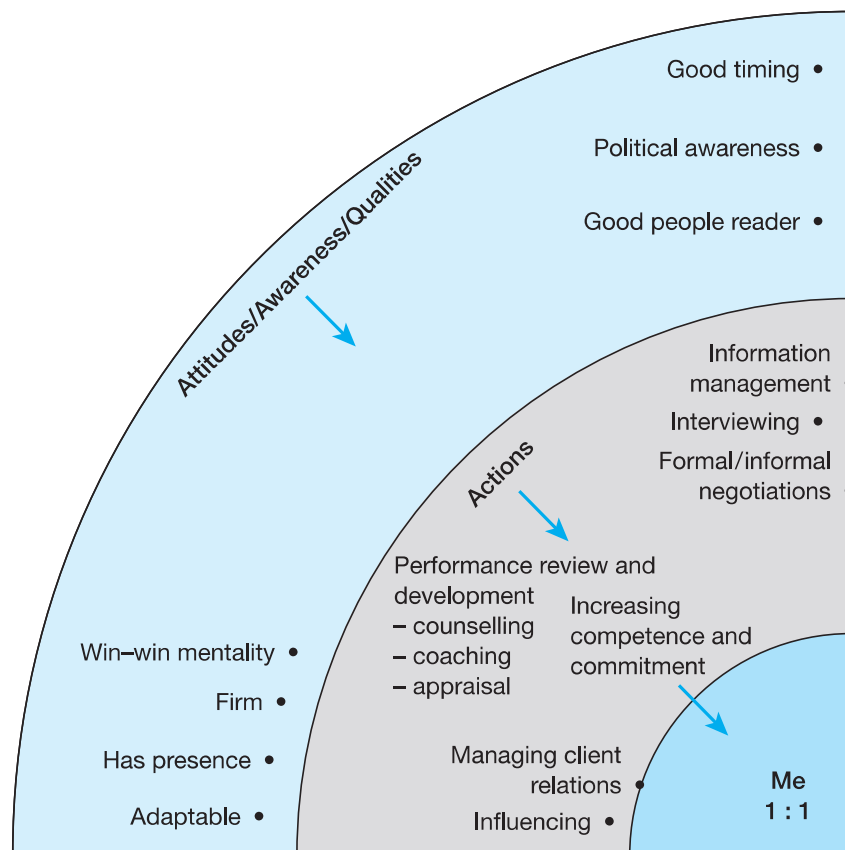


Figure 23.2 One-to-one interactions

Project managers frequently need to deal with large groups of people, for example in meetings or when giving presentations. Many feel anxious at the thought of presenting to a potentially critical audience, but the skills involved in doing so are an important part of the role. Also, the project manager will want to support the team wherever possible, but must be able to be authoritarian and critical when necessary. A judicious mixture of praise and control will be appreciated by all the team. As the leader of the project, it is the project manager who will drive, guide and motivate the team. Doing this well will enable the project manager to command the respect of the team. Figure 23.3 shows many of these key qualities and skills.

Everything described so far is of no use at all if the project manager does not manage the task adequately. Good communication and excellent self-organization are no substitute for the planning, monitoring and control processes that the project manager will create. The effectiveness of these areas will be the key to the outcome of the project. So the project manager must be a capable planner, with the ability to identify the work to be done as well as the checks and balances needed to manage and control it. These key tasks are shown in Figure 23.4.

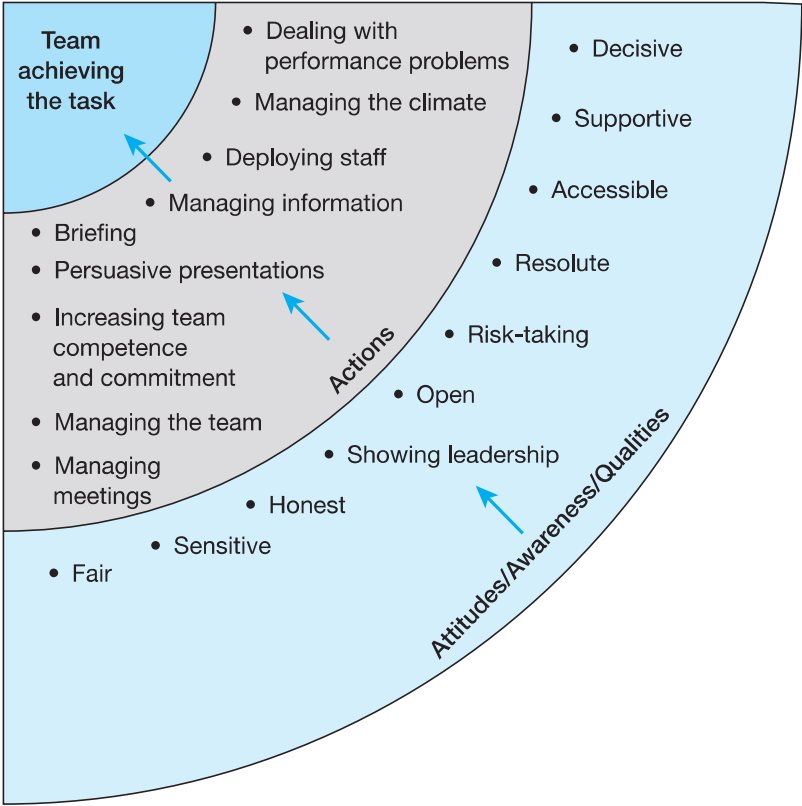


Figure 23.3 Managing others

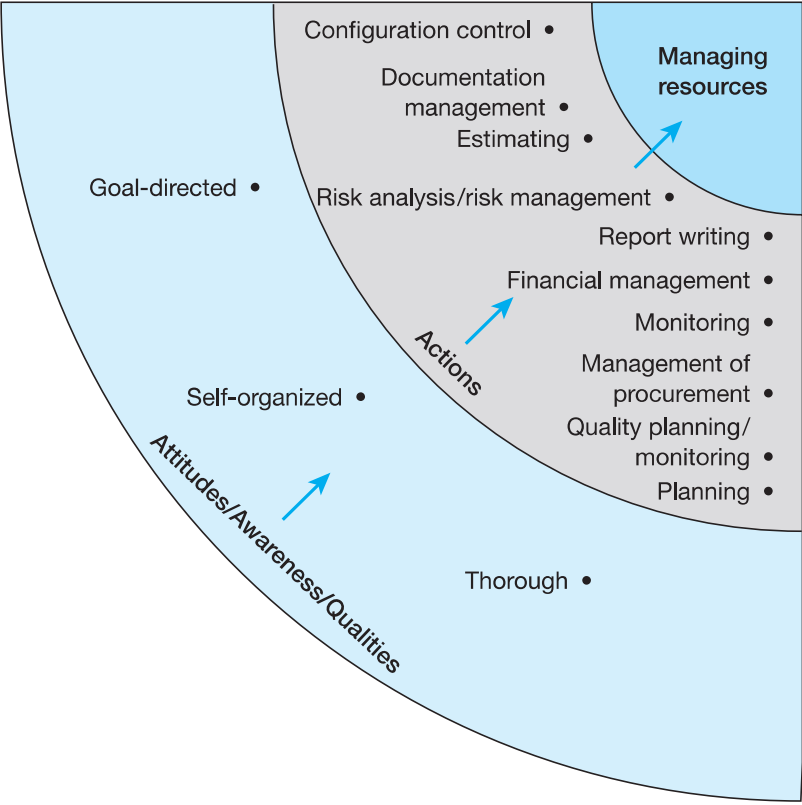


Figure 23.4 Managing resources

23.5 Using psychometric assessment

There is a truism: projects do not fail – people do. In companies that are project-led, nowhere is this more evident than in the selection and development of project managers. As will be obvious by now, a good project manager, well supported by more senior management, can mean the difference between the success and failure of the whole project. When people are most committed to the work they are doing and when they are most productive, there will be a number of factors at play. One of the most significant factors is that intangible one known as job satisfaction. For most of us this means a feeling of being challenged and of meeting the challenge; of working with others and helping them to give of their best as they help us to give of our best; of feeling positive about the work we are doing and the way we are doing that work.

The Institute of Personnel and Development published in early 1996 a study of staffing in UK companies. Its data gave turnover figures indicating that, on average, organizations can expect to replace the equivalent of their entire part-time staff every three to four years and to replace their full-time staff every six to seven years. If staff are unhappy, they will leave – if they can – more quickly than this. For each person who leaves, another has to be recruited, selected, trained and so on. The Institute's data demonstrated that the loss of just one professional – such as an experienced IS specialist – cost, on average, £23,000. And there are other, hidden costs too. The wrong person in the job will make mistakes while in post, is likely to miss opportunities, or perhaps commit the company to supplying something it cannot deliver. From this it is clear that:

- Companies need to ensure that they get the right people in the job as project manager.
- Companies must support, train and develop them on the job so that they perform well and have a satisfying job, ensuring that they want to stay with the company.

How can this be done easily and economically? Psychometrics can help, especially with selection and recruitment.

'Psychometrics' is often thought of as a buzzword, with many people using the results being unclear about what they really mean. However, with over 80 per cent of Britain's companies claiming to use psychometrics for recruitment, it is a good idea to outline their usefulness. Psychometrics are often called psychometric tests, as if somehow they could tell who were good candidates for a job and who were poor. It is more accurate to talk of psychometric measures or psychometric instruments, as many of them do not have passes or failures. They cannot distinguish between good and bad candidates – only a person can do that – but they can bring objectivity to the selection process. Correctly administered, there will be standardized conditions for carrying out the psychometric: standardized instructions, time, content, scoring and interpretation. This objectivity means that direct comparisons can be made, both with other candidates for a job and with an appropriate comparison group. And this means a greater likelihood of selecting the best candidate for the job.

There are three main types of psychometrics: measures of maximum performance or ability; measures of typical performance such as personality questionnaires; and interest inventories. Together these three can tell us a great deal about someone and provide information that it would be almost impossible to gain through interview. Let us now consider how they could be used to help in the selection and development of project managers.

Ability tests There are two main types of ability test (note that these are true tests, because it is possible to set a pass mark): attainment tests and aptitude tests. Attainment tests measure the results of formal training and are little used. Many employers simply check that a person has a particular educational qualification, and that can be done much more easily than by a test. Aptitude tests measure the ability of the person to acquire further knowledge or skills in a particular area. These are much more widely used. They are very good at measuring how good someone is in a particular skill area, such as verbal critical reasoning. If it is essential that the successful candidate is good at drawing inferences quickly and accurately from written text, then this is a good test to use.

Ability tests are not all paper-and-pencil tests. Simulations such as in-tray exercises are becoming increasingly popular, especially at senior management levels. Here the candidates are presented with a pile of letters, memos and items of background information. They are required to deal with this in a specified time and are awarded scores on abilities such as organizing, forecasting, decision making and written communication.

Interest inventories Interest inventories contain a wide sample of questions which might cover hobbies, school or university work, previous work experience or general life experiences and measure the direction in which an individual may want their career to go. Interest inventories are widely used in career counselling, and as such are particularly useful when someone is thinking of changing career – after all, that person may be able to do the job but would they enjoy doing it and would they stay in post? In both sales and management, for example, research evidence suggests that potentially effective managers may be identified by a specific pattern of likes and dislikes.

Personality questionnaires In many cases, the most important and useful type of psychometric measure is the personality questionnaire. Since personality is defined as ‘a person’s typical or preferred ways of behaving, thinking and feeling’, you can see how important personality is in determining how well or badly someone might do their job. If you were drawing up a list of the characteristics you would want in a good project manager, you may well come up with something like the one below which is used by a major airline:

- *Influence/leadership.* Able to take charge and control a group towards an objective. Makes their presence felt in groups. Motivates subordinates, delegates effectively and monitors performance.
- *Communication skills.* Able to comprehend and express ideas accurately and persuasively both orally and in writing.
- *Organizing/planning.* Able to think ahead, prioritize and organize work. Is able to structure activities and meet targets within appropriate budgets.

- *Motivation/energy*. Shows energy, drive and enthusiasm. Maintains a high output. Proactive and ambitious.
- *Creativity*. Generates and is receptive to new ideas. Seeks to innovate rather than necessarily accept the established solution.
- *Analytical*. Reasons objectively and critically with both verbal and numerical problems. Comprehends and processes information at a high level of complexity.
- *Empathy*. Able to understand the strengths, weaknesses, views and feelings of others. Supports others and is a good team member.
- *Emotional maturity*. Responds well to criticism. Frank and open. Able to take pressure.
- *Decision making*. Has the confidence to take own decisions. Is prepared to take a balanced risk and be held accountable for the outcome.
- *Commercial awareness*. An interest in, and understanding of, the financial and profit implications of actions taken. A hard-headed concern for the bottom line.

Look closely at each characteristic described here. How many of the qualities asked for are aspects of personality? In fact, every characteristic asked for in this list could be measured accurately by psychometric measures, leaving it up to the skilled interviewer to determine who would be the best person for the job from a range of people who could all do it well.

Research conducted with companies has established that up to 70 per cent of attributes associated with success at work are dimensions of personality rather than ability. Looked at in this way, it becomes vitally important to be able to measure these attributes in an objective and accurate way. From the point of view of the would-be project manager, the same considerations apply. Few people would want to pin their hopes of a successful career on something which would be unsuitable, so the earlier a person can discover a career which best suits their strengths, the better.

One particular company has used psychometrics to profile its project managers. Firstly, those managing project managers were asked to list the qualities that they greatly valued in the most effective of their project managers. The project managers were then 'tested' to see whether they exhibited what was sought by their bosses. The qualities that were most important to their managers were:

- Political awareness
- Being a good people reader
- Being good at negotiations
- Giving persuasive presentations
- Being good at influence
- Having a positive presence and confidence.

Next in order of importance were:

- Responsible leadership
- Risk-taking
- Innovative, problem-solving
- Strategic thinking
- Self-organized
- Goal-directed thoroughness.

Least important, though still of above-average importance, were:

- Assimilation skills
- Good decision making
- A win/win mentality
- Being supportive
- Being tough
- Being good at time management
- Being open-minded
- Sensitive.

The following list shows what was actually found. Each quality has a number beside it, between 1 and 10. An average score for a group of managers is 5, so we would expect all of the good project managers to be scoring 5 or more in each of the qualities their managers value. In the first group were:

- | | |
|---|---|
| ■ Political awareness | 5 |
| ■ Being a good people reader | 5 |
| ■ Being good at negotiations | 5 |
| ■ Giving persuasive presentations | 5 |
| ■ Being good at influence | 5 |
| ■ Having a positive presence and confidence | 5 |

Interestingly, although these qualities were the most important to their managers, the project managers were no better than average.

Next on the list were:

- | | |
|-------------------------------|---|
| ■ Responsible leadership | 8 |
| ■ Risk-taking | 3 |
| ■ Innovative, problem-solving | 6 |
| ■ Strategic thinking | 7 |
| ■ Self-organized | 7 |
| ■ Goal-directed thoroughness | 7 |

In this group, our project managers were mostly above average. The exception is in risk-taking. Further research indicated that the group of managers actually wanted what they called 'a safe pair of hands', someone who could be trusted with a major project. When they were initially talking about risk, what they really wanted was intellectual risk – innovation, in other words.

The least important list showed:

- | | |
|---------------------------------|---|
| ■ Assimilation skills | 9 |
| ■ Good decision making | 6 |
| ■ A win/win mentality | 6 |
| ■ Being supportive | 7 |
| ■ Being tough | 5 |
| ■ Being good at time-management | 4 |
| ■ Being open-minded | 6 |
| ■ Sensitive | 5 |

Here again we have almost the same picture. In many of these areas the project managers were above average. Two attributes are particularly worthy of mention. In assimilation skills, it was found that verbal skills were higher than mathematical skills, reflecting the fact that managing a project often involves having to read and understand long, complex documents.

In time management, what came to light was that many of the project managers – enough to skew the average score considerably – were highly involved in the detail of their projects. This frequently meant that they were unable to carry out all of their own work in the allotted time. All those who were asked about this admitted that it was a failing, but felt that the project could be at risk if they did not devote their attention to this level of detail.

What does this research tell us? It validates the perceptions of those who select, train and manage project managers in the company where the research was done. There was a company feeling that they knew what a good project manager was, and this was upheld by the tests: looking at the twenty key competences, the project managers came out below average in only three. In addition, this tells us that psychometrics measure what good project managers actually do, and can give us a yardstick to help establish whether someone new to the job would find it a satisfying career.

23.6 Ethical considerations

This section is not about the big business scandals that make the headlines – they are far beyond the scope of project management. There are, however, ethical issues that affect project management and, whilst we do not suggest that they need to be at the forefront of your mind, they have a real impact when they do occur. One part of the problem is that different ideas of right and wrong arise at different times and reflect changes in society. Another is that different cultures – national and organizational – have different views. We see that, for example, in civil law cases about racial and sexual discrimination. Ethical issues now occupy a more visible position in organizations. The general public, the media and special interest groups all focus more on ethical issues even though they may be shrouded in the cloak of scandal.

Increasingly, organizations are establishing their own codes of ethical conduct. These may be called ‘vision and value’ statements or describe ‘what we believe and how we work’, but they are nonetheless sets of moral guidelines that govern behaviour. Often they are based on the beliefs of the organization’s founder – Cadbury is an example – or on the purpose for which the organization was founded. The Protestant work ethic, based on the view that wealth is a sign of God’s favour, strongly influences many western organization cultures and hence their ethical stance.

But these complex issues, interesting though they are, lie outside the role of the project manager. There are, however, some ethical matters that lie within the scope of project management. There are two ethical frameworks that have an overriding influence. Put simply, they are:

- *Fundamental principle.* There are basic and unquestionable rights and wrongs. For example, theft and deceit are always wrong.
- *Consequences.* This is the 'ends justifies the means' view.

Let us take an example. Our project does not have enough copies of a software product. It is not very expensive, a few hundred pounds perhaps, but we are up against cost pressure and we need the software for a couple of weeks so that some extra staff can be used to meet a deadline. Do we make some pirate copies or not? Fundamental principle would lead us not to make copies because it would be theft. Consequences might allow us to justify it for the good of the project. However, in a 1992 analysis of 1,100 software piracy investigations, the Federation Against Software Theft reported that 21 per cent were the result of illegal copying by budget-conscious MIS managers (see Robson in the Further Reading).

So, what can project managers do? Based on an article in *Computing* in January 1995 by McNevin, some steps to follow when dealing with stakeholders where there is a contractual relationship, with supplies and customers for example, include:

- Defining the outcomes with absolute clarity.
- Keeping projects short and simple.
- Ensuring that suppliers are properly qualified and resourced.
- Ensuring that the project team is properly staffed.
- Dealing with differences over expectations as soon as they arise.

Project managers can also demonstrate their ethical standpoint and establish one for the project team through the use of the British Computer Society Code of Conduct outlined in Figure 23.5. The full Code is on the Society's website at www.bcs.org.uk. In the USA, the Association for Computing Machinery (ACM) has a code of Ethics and Professional Conduct at www.acm.org.

23.7 Summary

We have looked at the role of the project manager in a number of ways, and in each way the results have led to a similar conclusion. In one sense, we have gone full circle from the vision of the project manager at the start of this chapter to the list of qualities with numerical scales beside them which also describe a good project manager.

In each part of the chapter there is another important message: some things about the job can be learned – such as learning how to negotiate effectively, learning how to motivate the team or learning how to understand a contract – whereas other things are inbuilt. Anyone can learn the basic skills of managing a project; a few people, the right people, can go much further. And this is not just because they have the skills, it is also because they have the aptitude for career project management.

The Public Interest

This section deals with members' responsibility for public health and safety issues, the protection of the environment and the safeguarding of basic human rights. Members are to ensure that they have knowledge and understanding of relevant legislation and that they comply with it.

Employers and Clients

Members are expected to work with due care and diligence and deliver work on time and to budget and to report on any likely overruns as soon as practicable. If their professional judgement is overruled they are to point out the likely consequences. Members are expected to observe the highest standards of financial probity and to respect client and employer confidentiality. There should be no misrepresentation of the capability of products or services.

The Profession

This section of the Code requires members to uphold the reputation of the profession, to work to professional standards and to avoid any action that will adversely affect the good standing of the Profession. There is a responsibility to represent the Profession in public in an honest and balanced way and to act with integrity towards fellow members and members of other professions. Members are expected to encourage and support fellow members in their professional development.

Professional Competence and Integrity

Members shall continue to develop their professional competence and seek to conform to good practice. They must offer to do only that work that is in their area of competence and to accept professional responsibility for it and for the work of their subordinates.

Figure 23.5 Outline of the British Computer Society Code of Conduct

Questions

- 1 How does the 'vision of the project manager' in this chapter relate to the way you see the job? Are there aspects of the job that do not appear in the vision? Why might that be?
- 2 Consider the skills and qualities of project managers described in the 'developmental approach' (section 23.4). Can you add to these? How far do you see yourself being proficient in these skills? How could you develop further?

Further reading

- Cleland, D I and King, W R (1988), *Project Management Handbook*, Wiley
- Gaddis, P (1960), 'The project manager', *Harvard Business Review*, May/June
- Robson, W (1997), *Strategic Management and Information Systems*, 2nd edn, Prentice Hall
- Turner, R J and Simister, S (2000), *The Gower Handbook of Project Management*, 3rd Edition, Gower